Scaling in The Cloud

Cloud Computing

2026

Reference

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- Jamsa, K. (2022). Cloud computing. Jones & Bartlett Learning.

Overview

- Resource virtualization technique creates room for the adoption of dynamic approach in computing resource provisioning.
- The dynamic resource provisioning approach in turn creates the scope for developing scalable computing systems and applications.
- Scalability of systems and applications is an essential feature of cloud computing.

What is Scaling?

- Scaling is the characteristic of a system, model or function that describes its ability of growing or shrinking whenever required.
- In computing, scaling represents the capability of a system or application to deal with varying workload efficiently without bringing in a situation where resource shortage hampers performance or resource surplus increases the computation cost.
- In simple words, scaling is defined as the ability of being enlarged (or shrunk) for accommodating growth (or fall-off) to fulfill the business needs.

Scalability of System

- A system or application architecture can be termed as scalable if its performance improves on adding new resources and the improvement is proportional to the capacity added.
- The scalability of a scalable system is measured by the maximum workload it can competently handle at any moment.
- The point at which a system or application can not handle additional workload efficiently, anymore, is known as its limit of scalability.
- Scalability reaches its limit when a system’s architecture does not support scaling anymore or some critical hardware resource run out.

Scaling in Traditional Computing

- In traditional computing environment, once the resource capacity of a system is enhanced manually, the status of the system is retained till further human intervention, even if those resources do not get utilized.
- This under-utilization causes wastage of resources and increases the cost of computing.
- The main reason behind this problem is that the infrastructure architecture is not dynamic in traditional computing system.
- Static scaling requires system shut-down (system to restart) and hence is avoided unless it becomes extremely essential.

Scaling in Cloud Computing

- Scaling is one of the attractive attributes of cloud computing.
- In dynamic-automatic scaling, the system resource capacity can be altered while a system is running.
- Large pool of virtualized resources promises to adjust variable workload by allowing optimum resource utilization.
- The scalability in cloud computing is offered in a transparent manner and the automatic scaling ability gives the impression of infinite resources to its consumers.

Benefits of Scaling in Cloud Computing

- The scaling ability provided by cloud computing is particularly helpful for applications and services to meet unpredictable business demand.
- Resources in cloud can be acquired in seconds through the API calls.
- Cloud system architecture's ability to handle a heterogeneous pool of commodity hardware components has contributed to its infinite scalability.
- The loosely coupled components under this architecture can scale independently of each other while enabling the system to scale to extraordinary levels.

Scaling in Cloud is Reversible

- Virtualization has introduced the idea of dynamic infrastructure which has ignited dynamic resource provisioning.
- Dynamic scaling of computing system has become a reality through the dynamic resource provisioning approach where a system can be scaled without being compelled to restart.
- Thus, addition and removal of resource capacity to and from a running system has become viable and it has become much simpler to exploit extra capacity in cloud computing.
- Any time during the running of a system, extra resource capacity can be given back to the resource pool and can be claimed again whenever it is required.

Foundation Of Cloud Scaling

- Scaling in cloud is dynamic and automatic.
- Auto scaling feature in cloud computing has been achieved based on three pillars:
  - Resource virtualization: It eases resource management tasks and reduces the complexity of system development.
  - Resource sharing: It allows the optimal resource utilization by sharing resources among multiple users as well as applications.
  - Dynamic resource provisioning: It supplies (or reclaims) the resources on-demand, in response to the increase (or, decrease) in workload

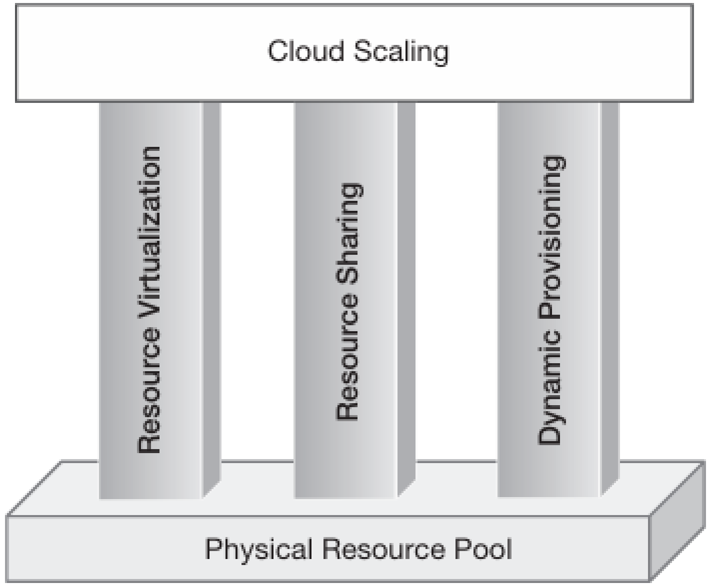

FIG 9.1: Foundations of cloud scaling

Classification

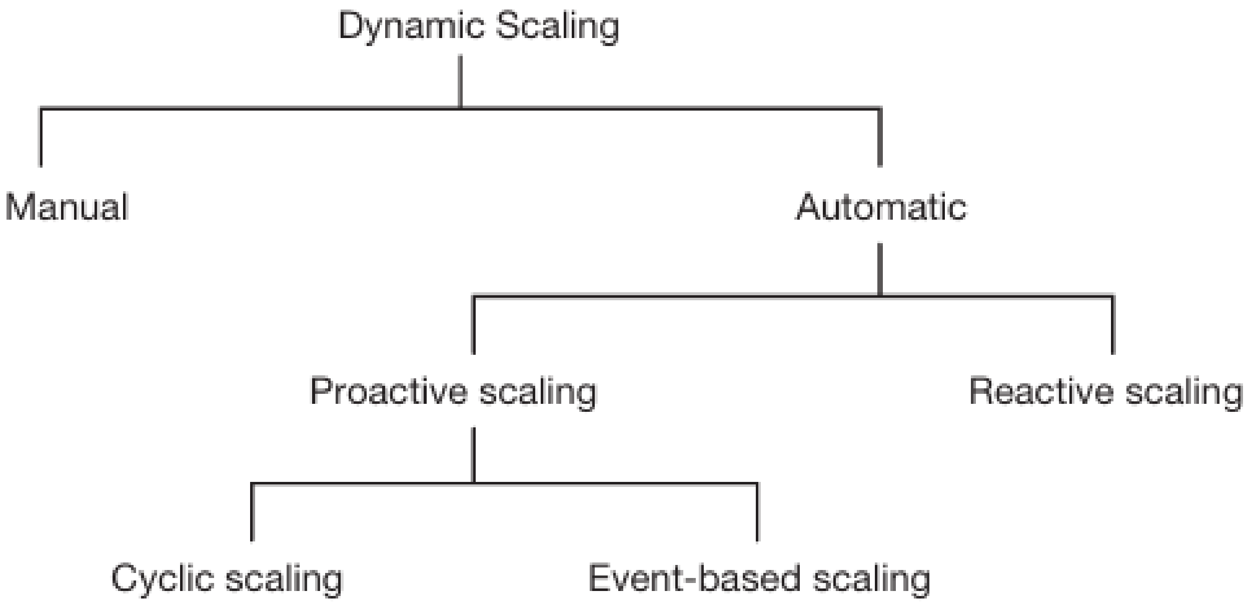

FIG 9.2: Classification of dynamic scaling approaches

Vertical Scaling

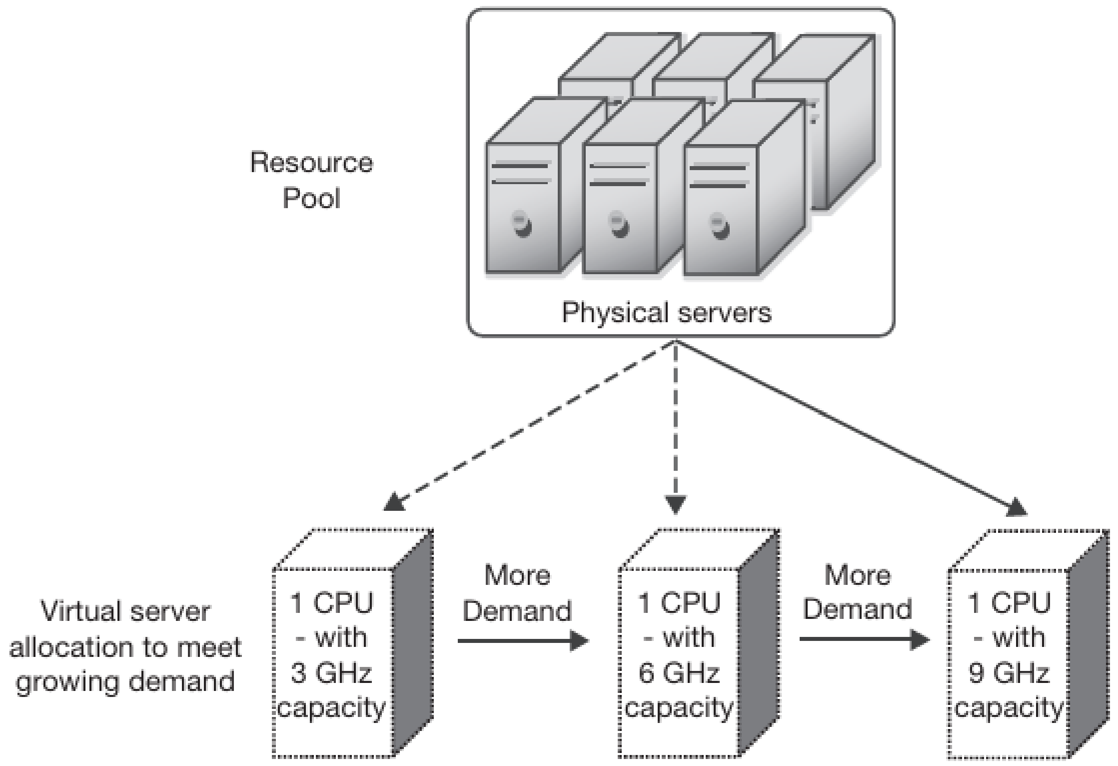

FIG 9.4: Computing resource scaled up by replacing it with more powerful resource as demand increases

Horizontal Scaling

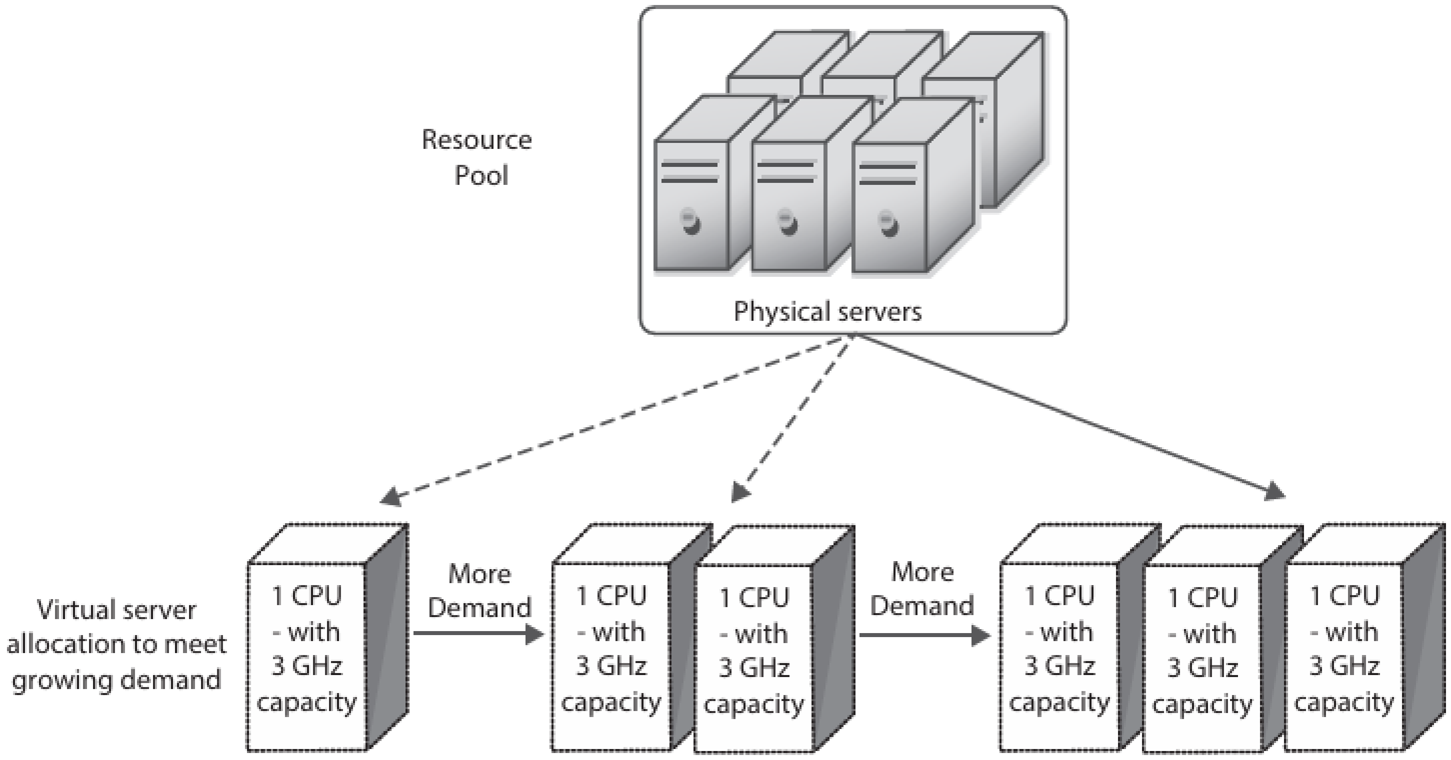

FIG 9.5: Computing resource scaled out by adding more number of same resources as demand increases